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Black box models vs White box models

The difference between black-box models and white-box models lies primarily in their interpretability and transparency when it comes to understanding how they make decisions.

General Comparison

Feature	**Black-Box Models**	**White-Box Models**
Transparency	Low (opaque)	High (transparent)
Interpretability	Hard to understand	Easy to understand
Common Examples	Deep Learning, Random Forest, XGBoost, Nonlinear SVM	Linear regression, simple decision trees, logic rules
Accuracy	Typically high (in complex tasks)	Moderate (but sufficient in many cases)
Use in critical environments	Limited without added explainability	Preferred when legal or ethical traceability is required
Post-hoc explainability	Requires tools like SHAP, LIME	Often not needed

Cuadro 1: General comparison between black-box and white-box models.

What Are They?

Black-Box Models

- The internal decision process is not easily understandable by humans.
- These are powerful, nonlinear models capable of learning complex patterns.
- They often require external methods to explain predictions.

  <u>Examples:</u>

  - Deep neural networks (CNNs, RNNs, Transformers)

- Ensemble models like Random Forest and XGBoost

White-Box Models

- Their internal logic is transparent and traceable.
- You can directly inspect weights, coefficients, or decision rules.
- Useful in domains where explanation is as important as prediction.

  Examples:

  - Linear or logistic regression
  - Simple decision trees
  - Rule-based expert systems

When to Use Each?

Context	**Recommendation**
Interpretability is critical	White-box
High accuracy needed for complex tasks	Black-box (with post-hoc explainability)
Regulated domains (healthcare, finance)	White-box or black-box with SHAP/LIME
Fast prototyping / exploration	White-box (simpler to tune and debug)

Cuadro 2: Recommendations for choosing between black-box and white-box models based on context.
